
Explainable AI:

A comparison between different methods on CUB200

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Abstract

Our project focuses on a comparative analysis of different explainable AI methods applied to a classification task on a fine grained dataset such as CUB200 using a Convolutional Neural Network (CNN), which in this case will be ResNet-50. The main goal is to address the "plausibility" and partially the "faithfulness" of the model explanations, this is possible thanks to the prior annotations on the dataset that allow us to visualize and analyze various explainability methods imported from the Captum library, an open source model interpretability for PyTorch. The main results report the degree of alignment of the model explanation with the annotations of the ground truth available in the dataset.

1. Introduction

Explainability in the context of Artificial Intelligence and Machine Learning is a fairly recent concept that started a new branch which is in continuous evolution.

The necessity to study this concept arises from the need to understand and give clear explanations to the results of models that continue to have an internal functioning which is less and less interpretable by humans as time passes. The latest architectures such as Transformers, but even the less recent ones such as Random Forest, constitute a clear example of this problem, which arises conflicts especially when it comes to high stakes applications where an explanation for a diagnosis or for a loan rejection is a right for those who want to receive it.

Among the several issues encountered through the years trying to solve this problem one of them seemed to be to decide what classifies as an explanation, this work will mostly focus on "plausibility", which can be synthesized as how convincing or reasonable an AI explanation appears to humans. This metric is typically assessed by comparing the algorithm explanation to human prior knowledge or expert annotations, as it is possible to do with the chosen dataset for the project (CUB200).

The chosen high level approach it is in fact to compare how plausible an explanation is through the lenses of different methods, that make use of the dataset and ResNet50 properties, to produce images that will be compared and analyzed with literature based evaluation metrics.

2. Related Work

For what it regards the gradual progress in this field of study and in particular the main topics covered in the project, it is fundamental to start from the first studies to continue on following both a topic and chronological order as this allows for a clear overview of the innovations and choices made through recent years.

A foundational study in model visualization and interpretability is "Deep inside convolutional networks" by (Simonyan et al., 2014). This work attempts to open the "black box" and to use the backpropagation mechanism, calculating the derivative of the output class with respect to the input, to create a saliency maps. It also allows for a clear inspection of what the model sees, initiating a discussion on the fact that the network is not in fact memorizing noise but learning shapes. This line of work is continued by Sundararajan et al. whose work implements integrated gradients, a method that attributes the prediction of a deep network to its inputs by summing gradients along a path from a noisy image to the real input. (Sundararajan et al., 2017)

The attempt to open the inside of a model is not the only way to get an explanation, this is clearly demonstrated by "Why Should I Trust You?", (Ribeiro et al., 2016) this work ignores the inside of the model and tries to disturb the input, observes how the output changes, and fits a simple explainable linear model locally. This innovation argues that "trust is crucial for effective human interaction with machine learning systems, and that explaining individual predictions is important in assessing trust". The new method proposed is LIME (Local Interpretable Model Agnostic Explanations), a modular and extensible approach to explain the predictions of any model in an interpretable manner.

Furthermore, trying to assess the situation of this field of study Lundberg et al. express that "various methods have recently been proposed to help users interpret the predic-

tions of complex models, but it is often unclear how these methods are related and when one method is preferable over another.” To address this problem they (Lundberg & Lee, 2017) present a unified framework for interpreting predictions, SHAP (SHapley Additive exPlanations). This work leverages Game Theory in a way such that each feature in the dataset is treated as a ”player” in a game, the model’s prediction is the ”payout,” and the Shapley value quantifies how much each feature contributed to the prediction compared to the average prediction.

All these methods can produce results that are visually compelling but this does not mean that they are also inherently reliable. This concept is argued by Adebayo et al. whose work try to give researchers guidance in assessing the scope of model explanation methods envisioning these methods to serve as sanity checks in the design of new model explanations. The results show that visual inspection can lack sensitivity to the model and the data generating process. *”Invariances in explanation methods give a concrete way to rule out the adequacy of the method for certain tasks”*, they primarily focused on invariance under model randomization, and label randomization (Adebayo et al., 2018).

Taking into account these limitations of purely visual explanations and as time passes and new architectures arise, it is natural to wonder how the evolution of models has also influenced interpretability techniques. In particular, with the advent of Transformers and LLM the focus has shifted from saliency maps to natural language explanations and mechanistic interpretability which is covered in the work of (Wei et al., 2022), an interesting path that goes out of the scope of this project.

3. Methodology

Once we assessed the high level approach and the literature based works around the main project idea, in this section we will provide a more in depth explanation of the methods used and the details of the work, all of this while underlining the reasons behind the different choices made.

3.1. Architecture and Techniques

As the main focus regards the visual explanations from a CNN, for the classification task we chose to implement a pipeline based on ResNet-50, our choice is mainly based on the fact that it is a standard and documented architecture, preventing structural complexities from obscuring the analysis of the explanations. ResNet has a 50-layer network, designed to overcome the vanishing gradient problem in very deep networks through the use of residual connections. Some features of this architecture are the Input Size which is 224×224×3, the Pretrained Weights on ImageNet (1000 classes) and the fact that is commonly used for fine tuning,

Table 1. Explainability methods and main attribution formulations.

	Method	Formulation
Grad.	Saliency	$ \partial f(x)/\partial x $
	Input×Grad	$x \odot \partial f(x)/\partial x$
	Int. Gradients	$(x - x') \int_0^1 \partial f(x' + \alpha(x - x'))/\partial x d\alpha$
Pert.	LIME	$\operatorname{argmin}_{g \in \mathcal{G}} \frac{1}{m} \sum_{i \in [m]} k(\mathbf{x}, \mathbf{x}_i) \underbrace{L(g(\mathbf{x}_i), y_i)}_{\text{loss on } (\mathbf{x}_i, y_i)}$
	SHAP	$\phi(i) := \frac{1}{d!} \sum_{\pi} \Delta(i, S_{i,\pi})$
	Occlusion	$M_i = f(x) - f(x \setminus x_i)$

which will be introduced as a core aspect later on.

Table 1 reports the base attribution formulations of the considered explainability methods, while Noise Tunnel is employed as a stabilization technique on top of Saliency and Integrated Gradients.

After the decision about the architecture, it comes a crucial section which emphasizes the different techniques used to achieve explainable results.

We will divide these techniques in two main categories, as also the literature suggests, which are **Gradient Based methods** and **Perturbation Based methods**

Starting off from the first family of methods we can say that they calculate how much each pixel influences the output by looking at the gradients in the network. They are generally fast and accurate for localization.

- Saliency calculates the absolute gradient of the output with respect to the input. This method can suffer from gradient saturation: if the model is too confident, the resulting gradient can be zero even for very important features.
- Input × Gradient implements the multiplication of the gradient by the input itself. The main purpose is to reduce noise in black areas of the image such where input=0, the attribution becomes 0, thus improving visual cleanliness compared to pure Saliency.
- Integrated Gradients instead calculates the integral of the gradients along a linear path from a Baseline, which could be a black image, to the input image. For this method it is necessary to have a clear configuration that sets a number of steps to be followed. A good standard compromise between mathematical precision and speed.

For what it matters the Perturbation Based Methods they treat the model as a black box, blacking out parts of the image to see how the prediction changes.

- LIME (Local Interpretable Model agnostic Explanations) does not work on individual pixels but on "Superpixels". A common way to divide the image into a chosen number of segments is to use a SLIC (Simple Linear Iterative Clustering) algorithm. In practice this approach explains a prediction by fitting a local linear replacement model on perturbed samples obtained by masking superpixels in the image, using the replacement coefficients as attributions.
- The SHAP kernel calculates, for each feature, the average marginal contribution to the prediction considering all possible subsets of features so in practice, it measures how much the model's output changes when a feature is added to a set of other features, and then averages this effect across all possible combinations.
- Occlusion applies a sliding window over the image to cover local portions of the input and observe how the model's output changes, in this way, it identifies which regions of the image are most relevant and how their deletions affects the prediction, offering a very intuitive method for understanding the localization of visual information.

Finally, **Noise Tunnel** is an explanation stabilization technique that consists of adding noise to the input and repeatedly calculating the attribution map, then averaging the results obtained. This approach reduces noise and variability in saliency maps, making explanations more robust and readable; in our work, it is applied to both the Saliency method and Integrated Gradients.

3.2. Dataset

The dataset chosen for this project is **Caltech-UCSD Birds-200-2011 (CUB-200-2011)** (Wah et al., 2011) which is an extended version of the CUB-200 dataset, with roughly double the number of images per class and new part location annotations. The dataset contains 200 categories and 11.788 images, with annotations including 15 part locations, 312 binary attributes, and one bounding box per image.

This dataset was necessary for the final scope of this project due to its annotations which allow for a clearer evaluation based on what the explanation methods see.

Having determined the importance of the dataset we apply different transformations to the dataset's images, first of all employing a standard procedure for ImageNet-pretrained models like ResNet-50 resizing to 224x224 and normalizing with ImageNet statistics.

Finally a crucial step to achieve the final scope of the project is to create artificial ground truth masks as shown in Fig. 1, based on the annotation present in the dataset. This step has to be implemented because the pure coordinates would

not be useful to understand the model explanation, being a single point in the image. This procedure starts with the scaling of the coordinates for the new image size, then a circle shaped mask is drawn around the coordinate with a fixed radius of 10 in order for it to become helpful for the evaluation metrics.

4. Experimental Setup

Our full experimental pipeline consists on: Data preparation, model training and testing, explainability generation and evaluation.

4.1. Training Configuration

For the training configuration we chose to implement fine tuning pipeline based on ResNet-50. This means that instead of training the model from scratch we applied a fine tuning for all the models layers and parameters, starting from the base parameters of ResNet-50 trained on Imagenet-V2. This choice allows the model to specialize on the subjects of the dataset while starting from a good baseline. Another necessary change that was implemented is the modification of the last layer, bringing the model to classify 200 classes instead of 1000. In order for the model to have an optimal training strategy we decided to use modern and standard performance maximization techniques such as AdamOptimizer, CosineAnnealingR as scheduler and CrossEntropyLoss for multiclass classification.

As for what it regards the **fine tuning** pipeline for the images we also apply different data augmentation techniques to avoid overfitting in particular implementing, Random Horizontal Flip, Random Rotation (15 degrees max), Color Jitter and Normalization. For the **evaluation** part (test set) of the fine tuning, images go through only the deterministic operations necessary to pass through the model which are resize and normalization.

4.2. Explanations Generation

During the explanation generation phase we decided to test the methods on $NUM_SAMPLES = 300$ images, the main hyperparameters of the XAI methods were set as follows. For integrated gradient method, the number of steps for integration along the path was set to $n_steps = 25$. For the Noise Tunnel technique, the number of noisy samples generated for each input was set to $NT_SAMPLES = 5$. For perturbation based methods, the image was divided into a number of segments equal to 50 superpixels, while the number of perturbed samples used by LIME and Kernel SHAP was set to $N_SAMPLES_LIME = 500$ and $N_SAMPLES_SHAP = 500$. In order to ensure the reproducibility of the experiments, a fixed set of random seeds equal to 42 was used. All of these parameters were chosen representing a compromise

between approximation accuracy and computational cost.

4.3. Evaluation Metrics

This section presents the metrics used to evaluate the explanations generated by different XAI methods. These measures provide a quantitative reference for comparing the performance of the methods. Formulations shown in Table 2.

The first evaluation metric used is the **Energy Based Pointing Game (EBPG)**, this choice is based on the fact that the classic “Pointing Game” is too simple since it only asks whether the most important pixel falls within the ground truth mask. It is a binary test (yes/no). The EBPG solves this problem being not only interested in the peak but also in how much total energy/attention falls on the ground truth mask compared to the background. This is essential to understand whether the network has captured the entire shape or just a detail. We evaluate spatial accuracy by measuring the proportion of saliency magnitude that falls within the ground truth mask. Given an input x with bounding box B and a saliency map M , the metric is calculated as the ratio of the saliency inside the ground truth mask to the total saliency, where Ω represents the spatial domain of the entire image. This approach penalizes maps that, while identifying the correct peak, disperse attention on the background.

A useful metric is also **Area Under the Curve (AUC)**, where we treat the saliency map as if it were a binary pixel classifier. The AUC tells us how good the map is in general, regardless of how strong or weak the signal is. If a map has very low values but is well distributed over the ground truth mask, the AUC will be high (good), while the EBPG may be low. By defining the True Positive Rate (TPR) and False Positive Rate (FPR) as the intensity threshold varies, the metric is defined as the integral of the TPR with respect to the FPR. An AUC value of 1 indicates perfect separation between ground truth mask and background, regardless of the absolute magnitude of the map.

As introduced in the Related works (Adebayo et al., 2018) the **Sanity Check** introduces an important shift to analyze explainability methods, in particular we take the weights of the trained model and begin to progressively randomize starting from the last layer to the input layer. At each randomization step, the saliency map is recalculated and

compared to the original map using similarity metrics such as Spearman Rank Correlation. We expect a correlation close to zero. High correlation values would indicate that the explanation method acts independently of the model’s learning, for example by behaving like a simple edge detector.

Finally, we analyze the relationship between the quality of the explanation and the correctness of the prediction. We define the **Performance Gap** as the difference between the average value of the localization metric calculated on correct predictions and that calculated on incorrect predictions. A positive gap suggests that the model produces more focused explanations when the prediction is correct, providing a useful signal for diagnosing errors.

5. Results and Discussion

5.1. Main Results and Analysis

MODEL PERFORMANCE

We first report the classification performance of the fine tuned ResNet model on the considered dataset, in terms of Top-1 and Top-5 accuracy.

Metric	Value
Top-1 Accuracy	74.25%
Top-5 Accuracy	92.46%

Table 3. Classification performance of ResNet on the evaluated dataset.

QUALITATIVE RESULTS

Qualitative visualizations (Fig. 2) of the attribution maps are presented to provide an intuitive comparison of the different explanation methods and to highlight their localization behavior across samples. From the overall analysis of the attribution maps, we can see that gradient based methods tend to create sharper mappings. In particular, integrating the NoiseTunnel wrapper with the Saliency and Integrated Gradients methods allows us to obtain significantly sharper maps compared to the original methods. Conversely, explanations created using LIME and Kernel SHAP present a problem with regard to the number of segments, due to computational limitations, which results in a rough explanation.

PLAUSIBILITY EVALUATION

This subsection discusses the plausibility of the generated explanations, focusing on their alignment with human interpretable visual patterns and semantic regions of interest.

In this subsection, the plausibility of the generated explanations is analyzed, focusing on their interpretability and the

Table 2. Summary of Evaluation Metrics

Metric	Formulation
EBPG	$\frac{\sum_{(i,j) \in B} M_{i,j} }{\sum_{(i,j) \in \Omega} M_{i,j} }$
AUC	$\int_0^1 \text{TPR}(\tau) d\text{FPR}(\tau)$
Sanity Check	$\rho_s = \text{Corr}(M_\theta, M_{\text{rand}})$
Perf. Gap	$\mathbb{E}[S \text{Correct}] - \mathbb{E}[S \text{Error}]$

Method	EBPG	AUC	Perf. Gap
Saliency	0.22 ± 0.07	0.85 ± 0.07	+0.001
Input $\times$ Gradient	0.23 ± 0.08	0.82 ± 0.07	+0.010
Integrated Gradients	0.24 ± 0.08	0.84 ± 0.07	+0.012
LIME	0.17 ± 0.07	0.59 ± 0.17	-0.002
SHAP (Kernel)	0.11 ± 0.05	0.53 ± 0.16	+0.009
Occlusion	0.22 ± 0.09	0.80 ± 0.08	+0.025
IG + Noise Tunnel	0.26 ± 0.08	0.86 ± 0.06	+0.007
Saliency + Noise Tunnel	0.22 ± 0.07	0.89 ± 0.06	-0.001

Table 4. Quantitative comparison of explainability methods on CUB200, including Noise Tunnel variants. We report Mean $\pm$ Std for EBPG and AUC. The Performance Gap indicates the difference in EBPG between correct and incorrect predictions.

ability to highlight visual patterns and semantically significant regions. Quantitative results are reported in Table 4, Fig. 3, Fig. 4, Fig. 5 in terms of EBPG and AUC.

Overall, looking at mean results, gradient based methods show higher EBPG and AUC values than perturbation based approaches, suggesting a greater ability to target visually relevant regions of the image. In particular, Integrated Gradients performs better than Saliency and Input $\times$ Gradient. Applying Noise Tunnel to Integrated Gradients leads to further improvement in metrics, indicating what was also said in qualitative results.

Regarding perturbation based methods, Occlusion presents comparable performance to some gradient based approaches, probably due to its intuitive local masking mechanism, which allows for effective localization of relevant regions. In contrast, LIME and Kernel SHAP show lower EBPG and AUC values, highlighting a lower effectiveness in producing visually plausible explanations in the image domain. This result can be partially attributed also to the fact that some approaches such as LIME were originally designed for tabular data and their assumptions may be less suitable for high dimensional visual input.

The interpretation of the Performance Gap metric is referred to the section dedicated to Faithfulness Analysis, where it is discussed in relation to the consistency between the explanations generated and the actual behavior of the model.

FAITHFULNESS ANALYSIS

Unlike plausibility, which measures how correct the explanation is in relation to the ground truth, faithfulness is used to understand whether the explanation is honest with respect to the model. To measure this, we used metrics related to the EBPG method (Fig. 6), taking into account correct and incorrect predictions (Fig. 7) and the sanity check (Adebayo et al., 2018) as shown in Fig. 8, Fig. 9. Looking at the box plot for explanation quality, we can see that the average EBPG values for correct predictions tend to be slightly

higher than for incorrect predictions. We can see this distinction more accurately in the bar plot, which highlights performance gaps. Without a baseline, we can see that, on average, the performance gap is not very high, probably because the model locates useful parts even in incorrect classifications but, due to the strong interclass similarity, fails to correctly discriminate between species. With regard to sanity, we can see that, on average, when randomising the layers from the output to the input, the correlation between the predictions with the original model and with randomised weights tends to decrease.

5.2. Limitations and Negative Results

Although the results obtained highlight interesting differences between explainability methods, the work has some limitations. Firstly, perturbation methods, such as LIME and Kernel SHAP, did not achieve high performance, probably, as already pointed out in previous sections, due to the high computational cost, which limited the number of samples and the granularity of the perturbations applied. Finally, the assessment of the plausibility and fidelity of explanations remains partially subjective, and further quantitative and qualitative analyses on different datasets could provide a more complete view of the methods’ behavior.

6. Conclusion

In this work were compared different explainability methods applied to a ResNet-50 model for image classification of the CUB200 dataset. The results show that gradient based methods, and in particular Integrated Gradients combined with Noise Tunnel, produce more stable and visually plausible explanations, while perturbation methods are more sensitive to computational constraints and less effective on the image domain. Plausibility and fidelity analyses, supported by quantitative metrics such as EBPG, AUC and Performance Gap, provide useful indications on the quality of explanations and the differences between the various methodologies. This study therefore offers a reference for the choice of XAI methods in fine-grained image classification contexts and suggests possible future directions, including optimization of perturbation methods and extension to different datasets and models.

6.1. Technical Details

For all technical details, the full implementation, and usage instructions, please refer to the project’s GitHub repository and the README file. All scripts, dependencies, and configurations are fully described there for reproducibility. The repository is available at [Explainability-project](https://github.com/Explainability-project).

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¹GitHub repository

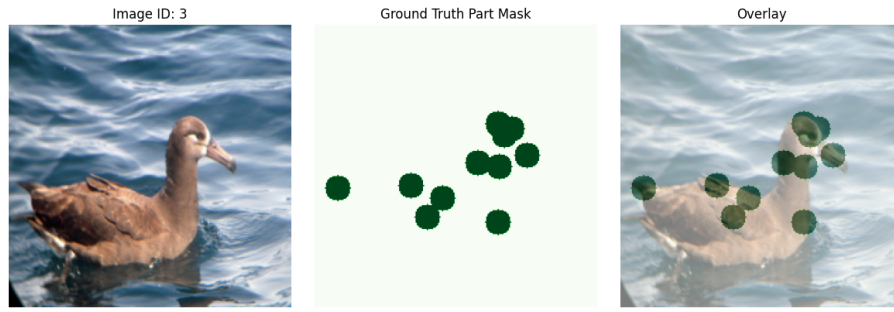


Figure 1. Ground-truth annotations used for the evaluation.

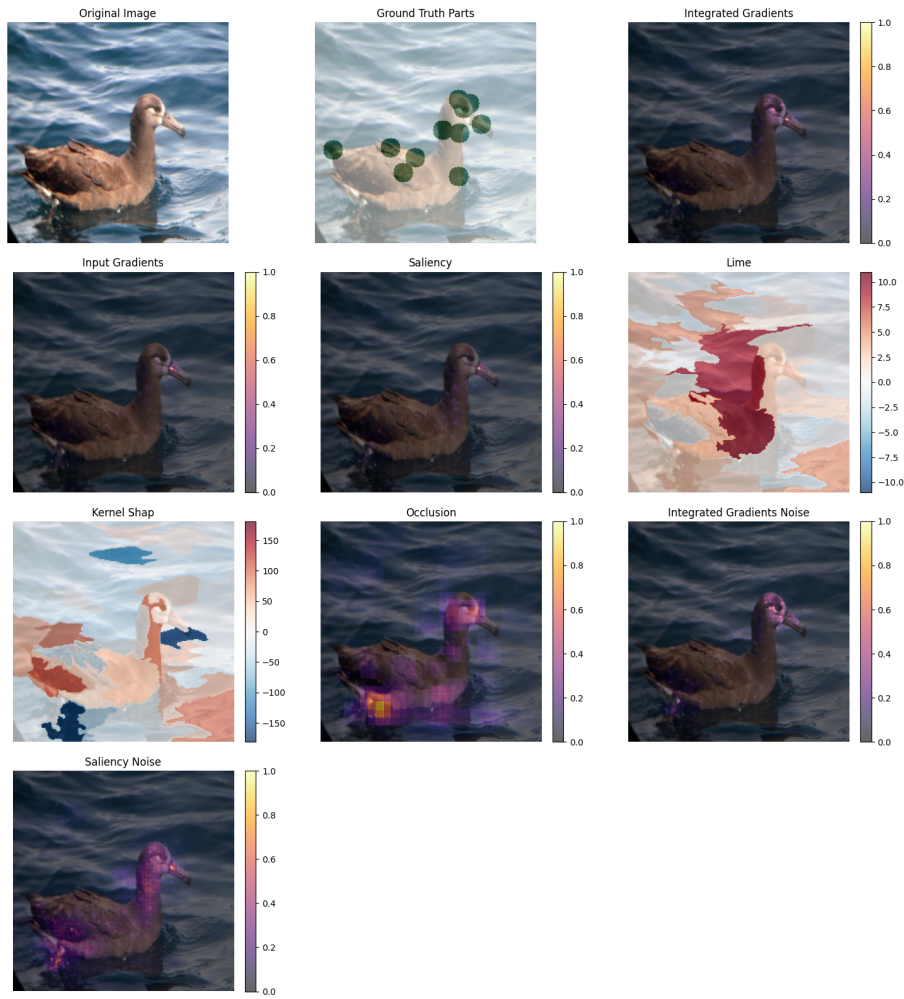


Figure 2. Example saliency maps produced by the considered explanation methods.

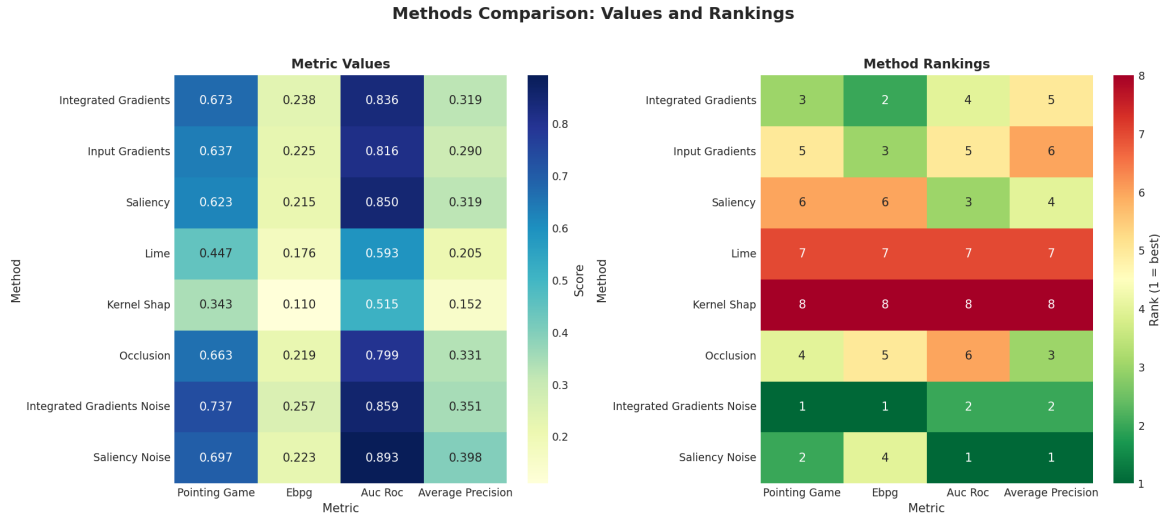


Figure 3. Heatmap showing the relative ranking of explanation methods across evaluation criteria.

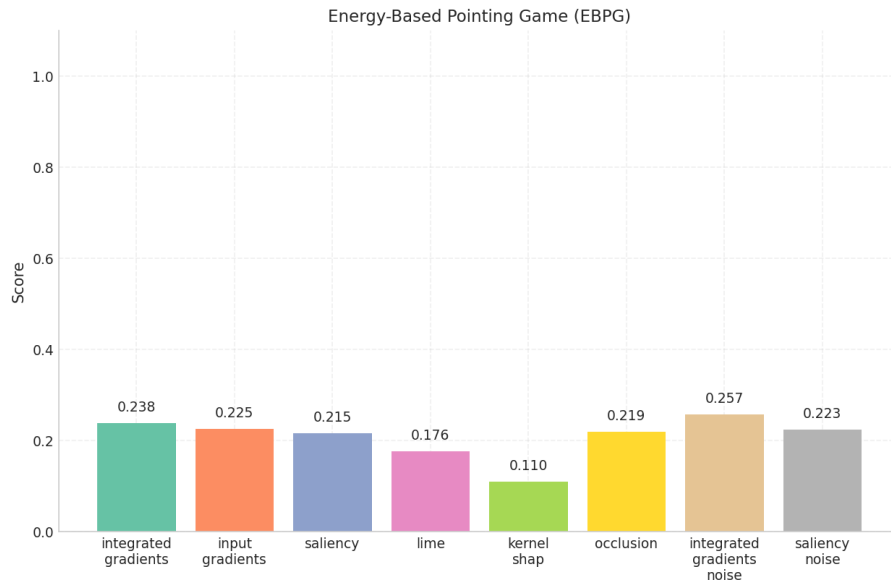


Figure 4. Comparison of methods in terms of EPG score.

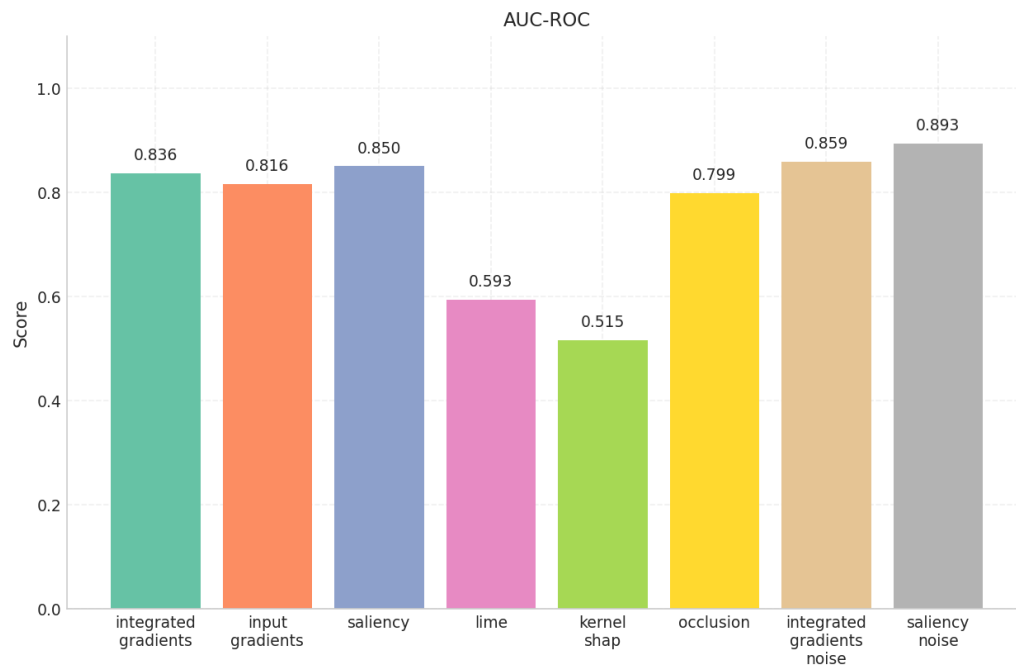


Figure 5. AUC-ROC performance comparison across explanation methods.

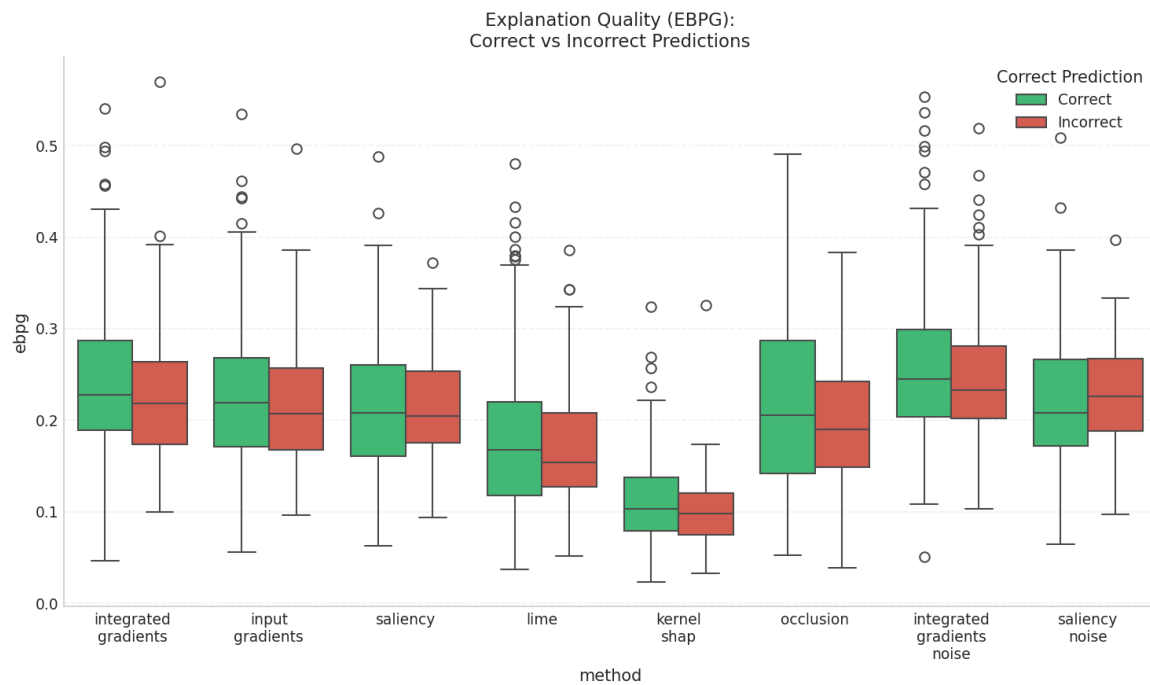


Figure 6. Model performance under different explanation-based settings.

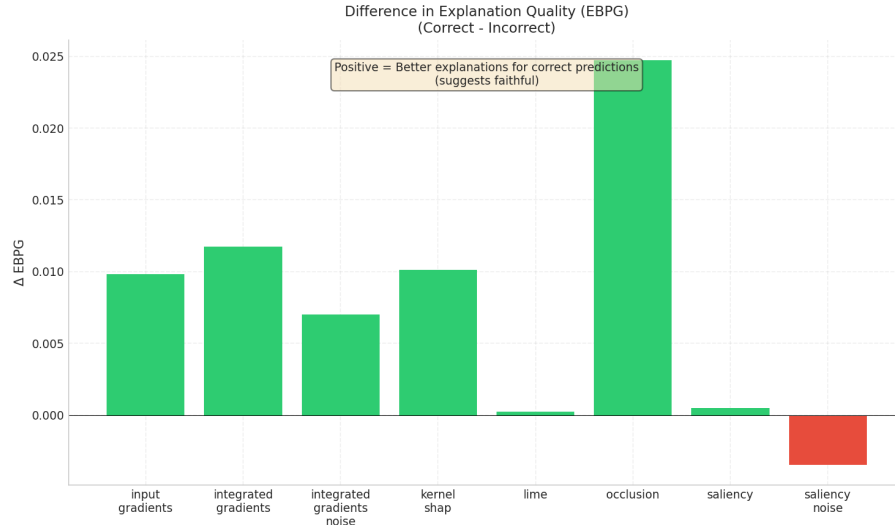


Figure 7. Performance gap induced by different explanation methods.

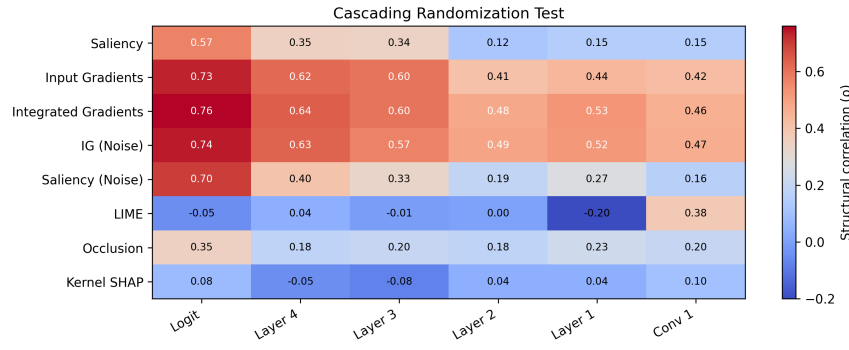


Figure 8. Cascading Randomization Test heatmap. Each cell reports the mean structural correlation (ρ) between the original explanation and the one obtained after randomizing a specific network layer.

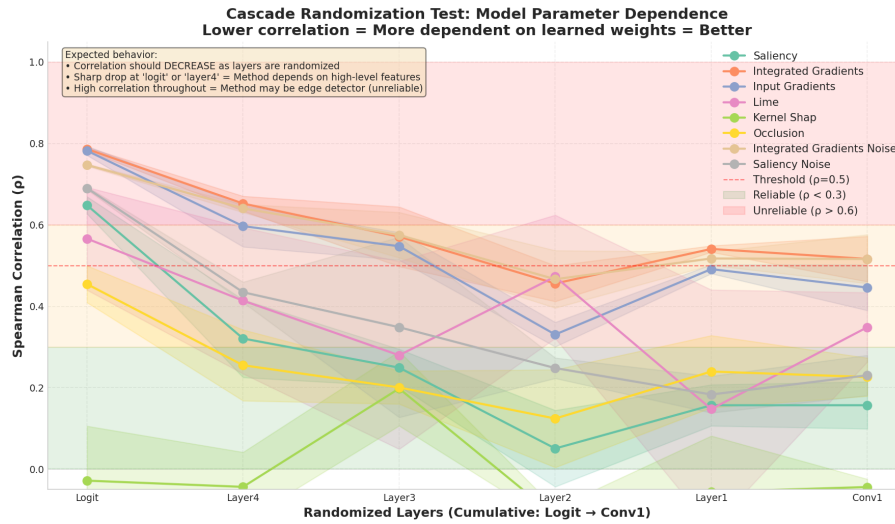


Figure 9. Cascading Randomization Test on CUB200. Saliency maps as model weights are progressively randomized from the output layer to early convolutional layers.